

## HETEROGENEITY AND DYNAMICS ON A SHOAL DURING SPRING -WINTER STORM SEASON, SOUTH-CENTRAL LOUISIANA, USA

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**Abstract:** Ship Shoal, a shore-parallel elongate sand shoal and a remnant of a late Holocene active delta has a unique heterogeneous sedimentary feature strongly affected by winter storms and fluvial sediment input from the Atchafalaya River. The interaction between fluvially derived sediment and subsequent deposition on the shoal has not been quantified; implications for hydrodynamic modeling are profound given that the shoal surface vacillates between sand and fluid mud. Thus, attenuation effects on waves and currents vary greatly. The results of a field survey undertaken during spring flood and winter storm periods showed that during fair weather, river-borne sediments transported to the shoal, forms a distinct fluid mud layer. Bottom sediments were re-suspended and transported by storm-induced waves and prevailing northerly/southerly

trends of currents over the shoal during the period. Data presented here suggest that the high sediment transport during the storms and the ephemeral deposition of fine-grained sediments could form transient sediment distribution on the shoal without leaving any modern sedimentary records, as reported by recent core surveys.

## INTRODUCTION

Much of the Louisiana continental shelf is characterized by fine-grained sediment due to fluvial inputs from the Mississippi and Atchafalaya rivers, their distributaries, and frequent hurricane and winter storms. Submerged transgressive sand shoals off Louisiana, which are remnants of past active deltas, form thick sandy sequences on the inner shelf (Penland et al., 1986).

Physical processes along the Louisiana coast are strongly affected by high energy events such as hurricanes and quasi-periodic winter storms. The muddy coast along western Louisiana also causes unique physical process responses (e.g. Sheremet and Stone 2003; Sheremet et al., 2005; Stone et al., 2006) as well as biological processes (e.g. Stone et al. 2006). Recent articles demonstrate the effects of the Mississippi and Atchafalaya River plumes on the Louisiana continental shelf, based on hydrological and geological surveys and satellite image analysis (e.g. Allison et al., 2000; Walker and Hammock, 2000; Bentley, 2002).

It is now well documented in the scientific literature that coastal Louisiana has experienced substantial wetland loss and beach/barrier island erosion due to high rates of subsidence, storms, shortage of sediment supply from adjacent fluvial systems and human intervention such as dam and levee construction. Federal and state governments are exploring potential sand sources offshore to restore eroding barrier islands and beaches (Drucker et al., 2004). Offshore sand resources have been considered one of the highest priorities for sand resources for coastal restoration given proximity to the target areas and for sand quality (Drucker, B., pers. com.). Ship Shoal, which is a remnant of the paleo-Maringouin delta, located 20 km offshore Isles Dernieres, is a high priority target sand resource to aid in the restoration of the Isles Dernieres Barrier Chain (Drucker et al., 2004). Numerous studies of the area have been undertaken particularly in terms of geological and physical processes (Penland et al., 1986; Pepper, 2000; Stone et al., 2006). Studies of the physical processes include wave-climate, bottom boundary layer, and sediment transport processes based on field surveys and spectral wave model studies (Pepper 2000; Stone et al. 2004). These studies concluded that waves substantially transform as they

propagate over the shoal by comparing the result of pre- and post-sand mining scenarios (Stone et al., 2004). Although they provide detailed mechanisms of hydrodynamics and non-cohesive sediment transport at the shoal, recent surveys suggest that the area has a complex sedimentary environment, which has not previously been researched. To fully understand the mechanism of wave-bottom sediment interactions and sediment transport processes, field surveys and numerical modeling studies were conducted and a synopsis of the data are presented here. The overall purpose of this paper is to describe wave and current dynamics affecting the bottom boundary layer, and sediment transport over the shoal during a field survey conducted in Spring 2006.

## DATA AQUITION

During spring flood and the winter storm season, an array of oceanographic instrumentation was deployed at three locations along the northern and southern

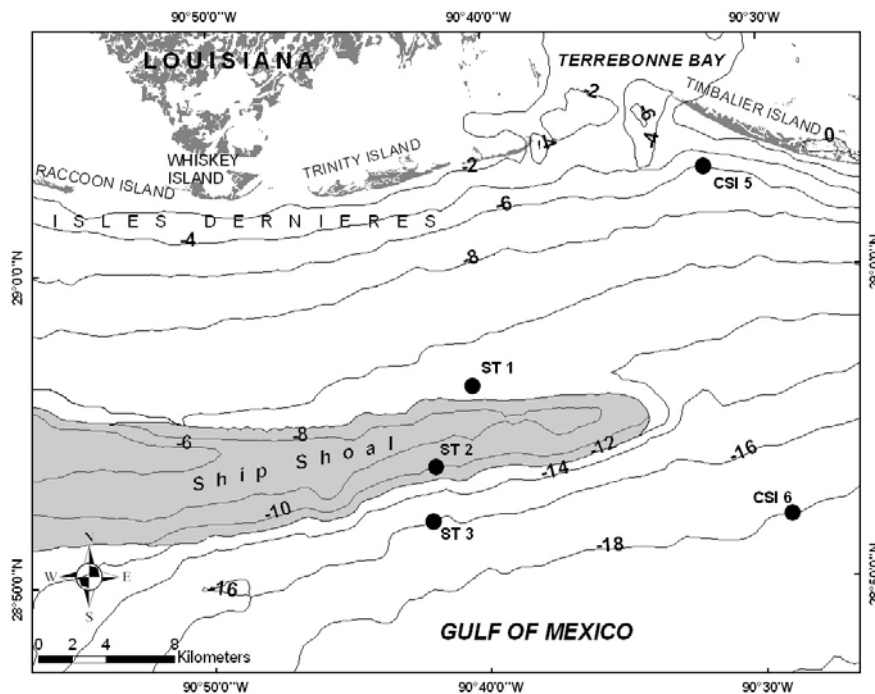


Figure 1 Map of 2006 survey stations, ST1, ST2 and ST3

edges of the shoal and the crest of the shoal, to collect wave, current and suspended sediment concentration data (Figure 1). The instruments at the middle station (ST2) consisted of a 1.5 MHz SonTek<sup>R</sup> pulse-coherent Doppler profiler, a Druck<sup>R</sup> pressure sensor, and two D&A<sup>R</sup> optical backscatter sensors. A 5.0 MHz SonTek<sup>R</sup> Acoustic Doppler Velocimeter and a ParoScientific<sup>R</sup> pressure sensor were deployed at both

edges of the shoal (ST1, ST3). Two D&A<sup>R</sup> optical backscatter sensors were also deployed at the north station (ST1). Each instrument system was tethered to nearby oil platforms for a robust deployment.

## DATA ANALYSIS

Significant wave height, peak wave period, and dominant wave direction were calculated from pressure and current data based on the spectral method by Earle (1995) (see also Stone et al. 2006). Directional wave spectra were calculated by the PUV method using DIWASP, a Matlab package (DIWASP, 2001).

Suspended sediment concentration (hereafter referred to as SSC) was estimated from the data collected using optical backscatter sensors (hereafter referred to as OBS) after calibration using water and sediment samples taken from the stations by ponar grab. The calibration studies were conducted at the CSI field support facility. Bottom sediment samples were incrementally added to distilled water in a black bucket and stirred using a hand-held electric agitator. OBS sensors were immersed in the bucket and measurements were taken for 2 minutes. Water samples were taken after the OBS measurements and were filtered; the SSC were estimated from the weight of the dried sediments and the water volume (Stone et al., 2006).

SSC was also estimated from the ADV and the PCADP single frequency acoustic backscatter intensity with appropriate calibration. The acoustic backscatter intensity is a by-product of the PCADP and the ADV, and is proportional to the SSC estimated from the OBS. This method includes inevitable noise because the backscattering is affected by various parameters such as sediment type, size and composition. All are difficult to quantify by single frequency backscatter sensors and there remain limitations associated with optical and acoustic instruments (Hamilton et al., 1998). In spite of these limitations, estimating the relative sediment concentration profile based on the acoustic backscatter intensity is a cost-effective method which enables us to monitor the SSC trend over time. A more detailed method of estimating the SSC from the OBS and the PCADP/ADV is presented in Stone et al. (2006). The SSC at ST3 was estimated from the relationship between the OBS and the ADV backscatter intensity at ST1.

Wave orbital velocity, shear velocity, shear stress, and apparent bottom roughness were calculated to examine boundary layer dynamics. For the calculation of shear velocity and shear stress, four methods were used: (i) linear wave theory, (ii) log-linear method, (iii) Reynolds stress method, and (iv) quadratic stress law method (Sternberg, 1972; Komar, 1976; Pepper, 2000). Critical shear stress for non-cohesive

sediments was calculated by the Shields parameter combined with the modified Yalin parameter (Pepper, 2000). The critical shear stress for cohesive sediments is difficult to specify and needs further laboratory experiment to determine parameters which are site-specific (Mehta et al., 1989). Mean critical shear stress for erosion and deposition is suggested to be  $0.15 \text{ N/m}^2$  and  $0.1 \text{ N/m}^2$ , respectively, based on a wide variety of studies in the U.S. and was used in this study (DHI, pers. com.).

## RESULTS AND DISCUSSIONS

### Meteorological and hydrological characteristics

Tropical cyclones (i.e. hurricanes and tropical storms) and extra-tropical storms (i.e. winter storms) are the two important meteorological events that affect the Louisiana coast. Particularly, frequent passage of winter storms between October and April causes cumulative effects on waves, hydrodynamics and sediment transport along the Louisiana coast because of their frequency, compared to less frequent landfall of tropical cyclones (Pepper, 2000; Stone et al., 2004; Kobashi et al., 2005). Generally, as cold fronts approach the coast, strong southerly winds can generate significant wave height on the shelf from 3-5 m, barometric pressure shows a marked decrease and thus characterizes the pre-frontal phase. After the front passes, air temperature abruptly decreases and wind direction changes from the southern to northern quadrants representing the post-frontal phase. The above described trend is evident in the time series of wind data obtained from a nearby meteorological station, WAVCIS CSI-6 ( $90^\circ 29' \text{ W}$ ,  $28^\circ 52' \text{ N}$ ) (Figure 2 left). The dominant wind direction is  $173$  degrees (i.e. wind from SSE), which is a common characteristic during the fair weather conditions and a force to generate longshore current, generally directed westward, off south-central Louisiana (Figure 2 right).

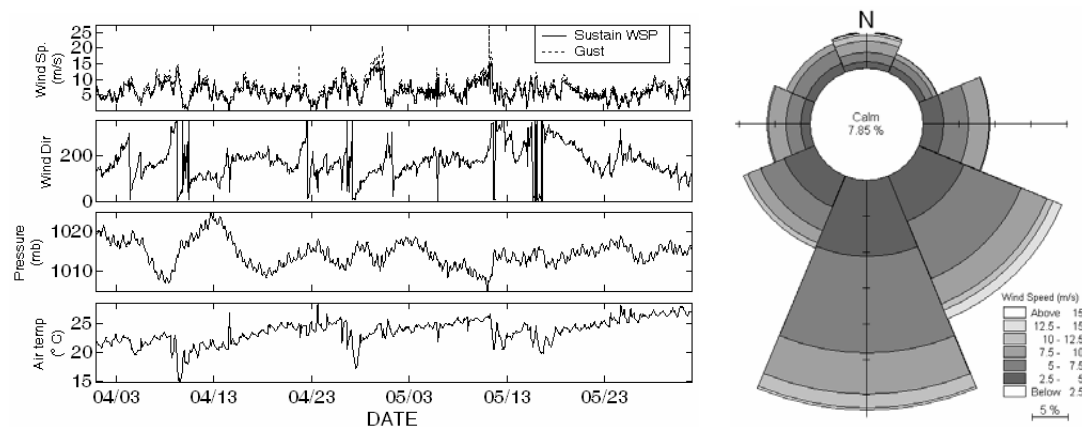


Figure 2 Time series of meteorological data (left) and wind rose (right) from WAVCIS CSI-6

Significant amounts of fluvial sediment discharged from both the Mississippi and Atchafalaya rivers, and distributaries, are also highly significant in introducing sediment to the shelf thereby playing an important role on hydrological processes that affect physical and biological processes on the Louisiana inner continental shelf. In the study area, maximum river discharges of 310,000 cubic feet per second (cfs) were recorded at Simmesport located upstream on the Atchafalaya River on March 30<sup>th</sup>, 2006 and 250,000 cfs on May 8<sup>th</sup>, 2006, resulting in significant influxes of muddy sediments along the western/south-central Louisiana coast (Stone et al., 2006).

### Bottom sediment characteristics

Bottom sediments collected during a biological cruise conducted for the entire shoal that included ponar grabs along the eastern flank of the shoal before and after bottom boundary layer equipment deployment (Figure 3, shaded triangles), show a unique shoal sedimentary environment. Bottom sediment grain size distribution for the entire shoal, surveyed in late May, 2006, shows that the grain size is finer along the western part of the shoal than its eastern flank; we attribute this to the influence of fluvially derived material from the Atchafalaya River (Figure 3, left).

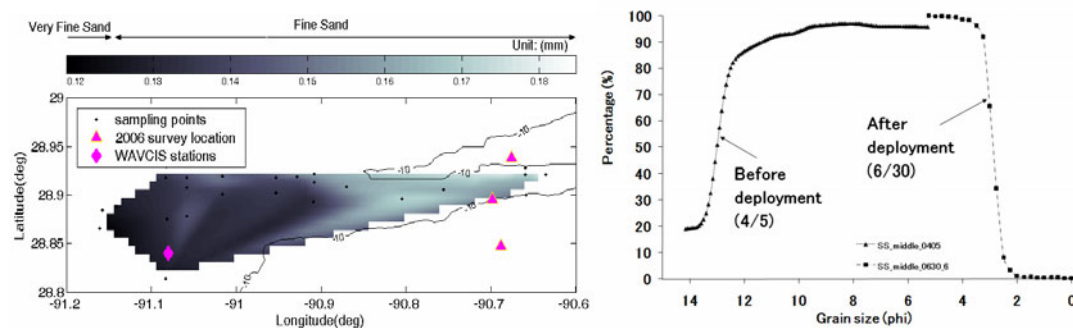


Figure 3 Mean grain diameter for the entire shoal (samples taken May 25<sup>th</sup>, 2006) (left) and grain size distribution taken before and after the deployment (right)

Bottom sediment taken from the eastern portion of the shoal shows a dramatic change in sediment grain size before and after the deployment (Figure 3, right). Sediment sampled before the deployment was mainly clay with a median grain diameter of 2 microns, probably originating from the Atchafalaya River; sediment sampled after the deployment was fine sand with a median grain diameter of 143 microns, suggesting the effect of sediment reworking due to storm-induced waves. This unique behavior of the bottom sediments was not found during previous field surveys undertaken along the western flank of Ship Shoal in January, 1998 (Pepper, 2000). The fine

grained sediments from the Atchafalaya River are usually transported west due to a longshore coastal current (Wells and Kemp, 1981). However, during winter storms, strong southerly wind-induced currents with sufficient sediment supply can transport this fine grained sediment to the south/southeast, occasionally reaching Ship Shoal (Kobashi et al., submitted). Such a sediment dispersal pattern was confirmed by satellite imagery obtained during our deployment and also reported by Walker and Hammock (2000).

### Wave characteristics

During the cold frontal passages, pre-and post-frontal phases have a significant effect on the wave field along the northern Gulf. Lower frequency waves dominate the spectrum during pre-frontal conditions. As winds veer to the north, higher frequency waves occur in the bays/estuaries and wave energy levels drop significantly on the shoreface (Figure 4d,e). Waves can become higher after the storms because of relatively strong southerly wind induced by a high pressure system covering the area (e.g. April 10<sup>th</sup>, 2006) (e.g. Pepper, 2000; Kobashi et al., 2005).

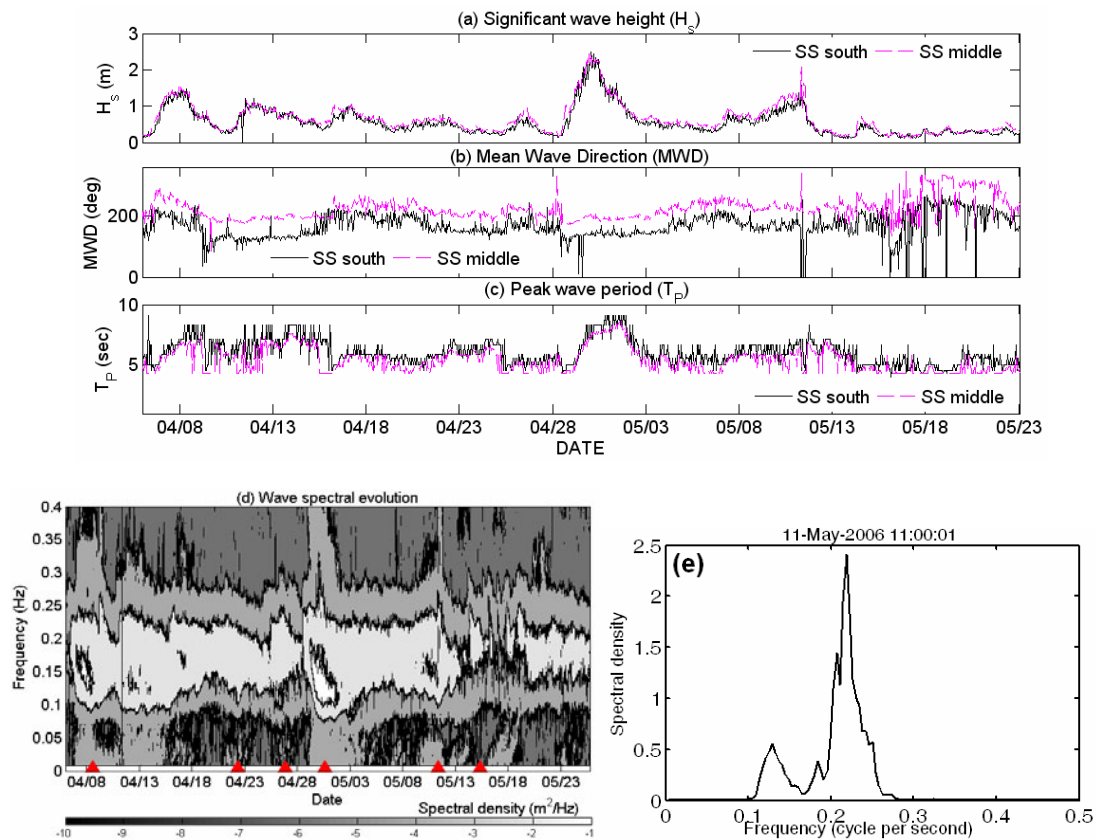


Figure 4 (a-c) Time series of wave parameters, (d) Wave spectral evolution, (e) spectral distribution during a post-front. Triangles show passage of cold fronts.

### Current velocity and circulation patterns

Bottom currents are strongly affected by local bathymetry and associated turbulence as well as currents due to tides. Current direction near the bottom is different from that at the surface, suggesting current modulation due to local bathymetry, wave-induced turbulence, or wind-induced Ekman transport. Currents due to tides are generally small (i.e. less than 5 cm/s) and strong currents occur during winter storms. The strong northeast/southwest oscillation of current direction during winter storms is clearly apparent in Figure 5, and is associated with the rotating storm wind direction. During fair weather, currents are weak and highly variable when compared to storm conditions; however, surface currents were often westward (not in the figure) due to longshore currents driven by prevailing south-easterly waves. In addition to the longshore currents, tidally-induced currents when combined with higher energy waves during winter storms result in additional currents with a period of around 5 to 10 days, a mean frequency consistent with winter storms.

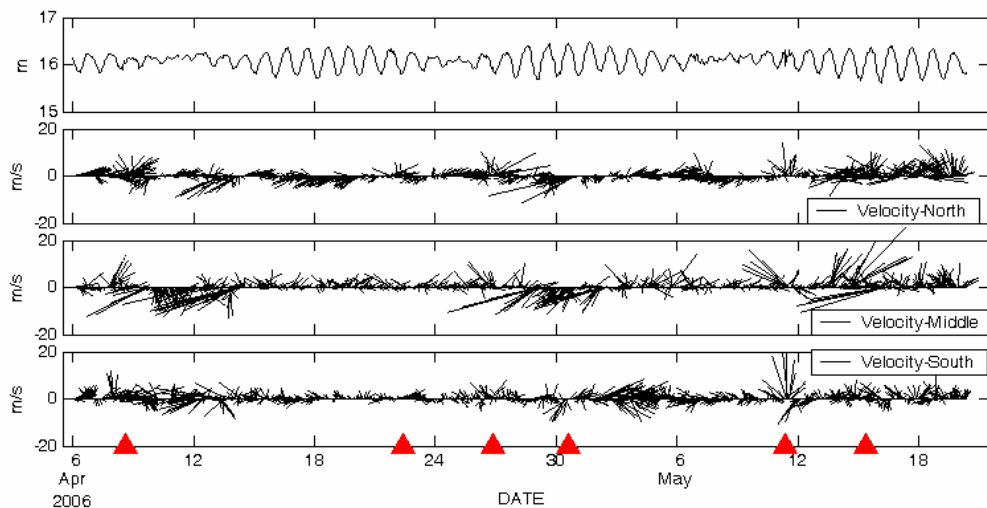


Figure 5 Stick vector of bottom current velocity at each station. Triangles indicate passage of storms

### Suspended Sediment Concentration

Suspended sediment concentration is also strongly associated with winter storms. Figure 6 shows that high sediment concentrations occur during winter storms at all three stations, suggesting that strong storm-induced waves cause high bottom sediment re-suspension compared to that during fair weather. Particularly, during late April, high concentration of suspended sediments associated with a storm occurred. At ST1, SSC attained saturation level and lasted approximately 2 days. At ST2 and ST3, high SSC also occurred during the waning phase of the storm as well as during



the pre-frontal stage, suggesting the occurrence of substantial deposition of cohesive sediments (The latter is discussed in more detail below). High SSC during April 8<sup>th</sup> and May 10<sup>th</sup> shows the input of river-borne sediments from the Atchafalaya River, also confirmed by satellite images and altimeter data during this period (Kobashi et al., submitted).

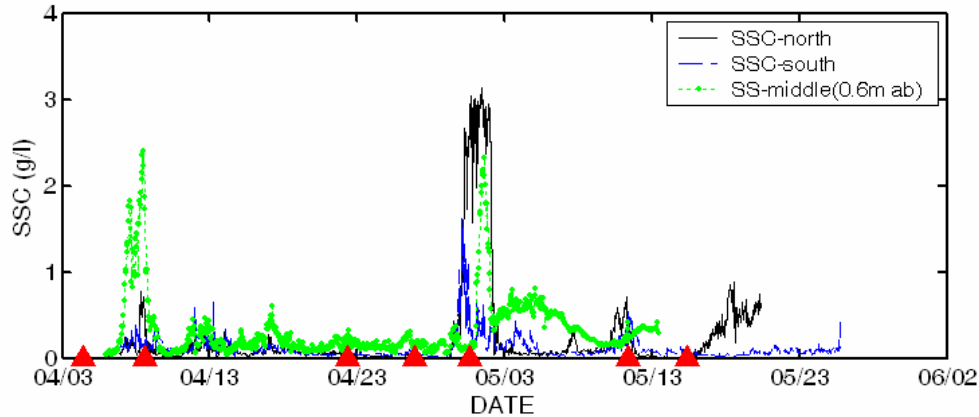


Figure 6 Time series of sediment concentration at each deployment station. Triangles indicate passage of storms.

### Bottom boundary layer parameters and sediment flux

Shear velocity and shear stress are high during winter storms. Bottom orbital velocities associated with winter storms attained a maximum of 60 cm/s during a storm in late April (Figure 7). The periods of high orbital velocity, shear velocity, and shear stress match those of the high SSC shown in Figure 6.

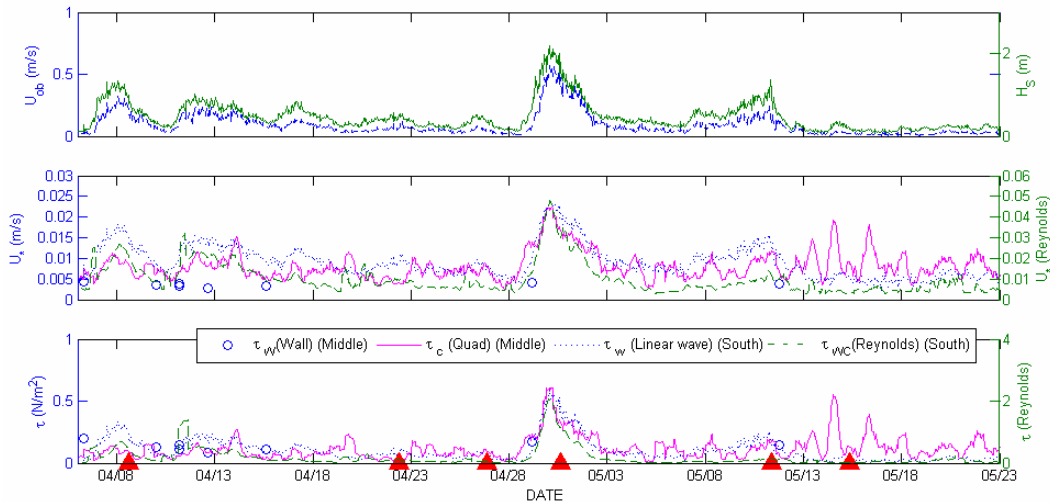


Figure 7 Time series of bottom boundary layer parameters: (a) wave orbital velocity ( $U_{ob}$ ), (b) shear velocity ( $U^*$ ), and (c) shear stress ( $\tau$ ) during the deployment at the middle station

The sediment flux, calculated by the product of horizontal velocity and sediment concentration, is shown in Figure 8. The direction of sediment transport is different among stations, and is significantly influenced by current direction. Northerly/southerly currents during winter storms drive the highest transport in the region, although current direction is dependent on other factors such as local bathymetry and tides. It is evident that in-situ observation of sediment flux is much higher than non-cohesive sediment transport derived numerically (not in the figure). It is also evident sediment flux occurs during fair weather, a finding that contradicts the results of previous work (e.g. Wright et al., 1997).

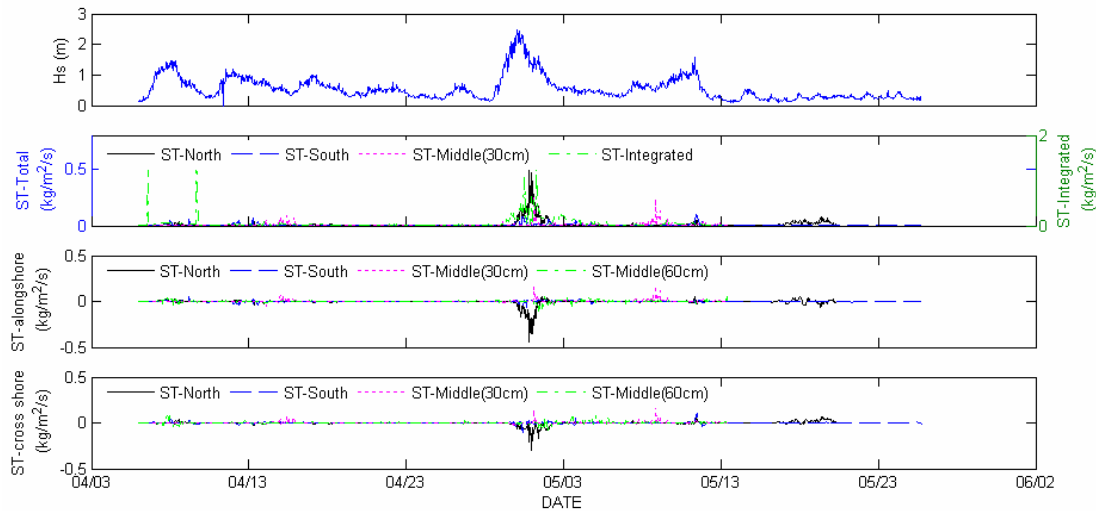


Figure 8 In-situ sediment transport measurements for each deployment station

### Resuspension, mixing, and settling processes of fluid mud during a storm

Waves, currents, and sediment transport are intensified by winter storms, and the bottom sediment of the shoal uniquely varies during these events. Figure 9 shows the time series variation of bottom boundary layer parameters and SSC during a storm which passed the region in late April, 2006. Wave height reached approximately 2.5 m during the pre-frontal phase, consistent with a maximum wind speed of 18 m/s. When wave height reached approximately 1 m, the shear stress exceeds the threshold for sediment re-suspension (i.e.  $\tau_{ce}$ ), which is consistent with the increase of bottom SSC (Figure 9c). Bottom sediments are re-suspended strongly during the pre-frontal phase (Re-suspension period), causing highest sediment transport during the storm because of high horizontal velocity. Subsequently, the transport rates decrease due to decreased horizontal velocity and an increased vertical velocity causing vertical mixing during the storm passage (Mixing period). During the waning phase of the storm, substantial sediment deposition occurs in spite of the high sediment transport

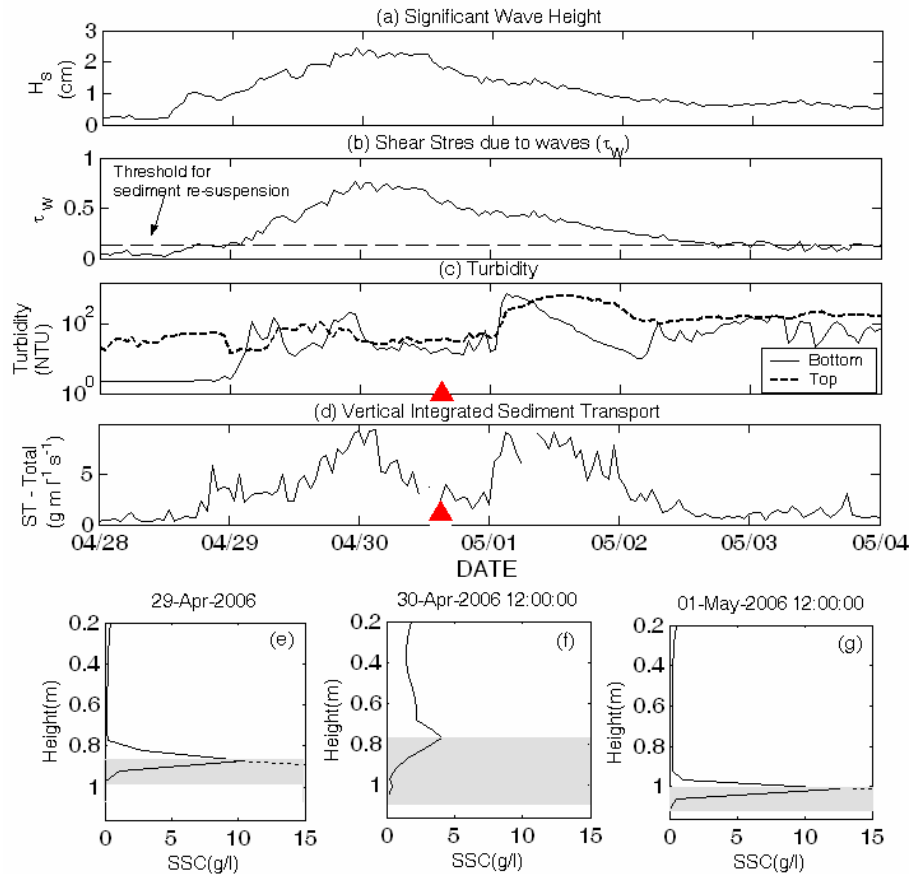


Figure 9 Time series of (a) significant wave height, (b) shear stress due to waves, (c) turbidity, (d) vertical integrated sediment transport, (e)-(g) sediment concentration profiles. Triangles indicate passage of a storm

rates (Settling period). The SSC profile in Figure 9 (e, f, g) demonstrates the above. Prior to storm passage, the peak sediment concentration was measured near the bottom, suggesting the existence of a fluid mud layer ( $SSC > 10$  g/l) delivered by hyperpycnal flows originated from the Atchafalaya River. The peak of the SSC strongly decreased below the lutocline since the acoustic signal was strongly attenuated below it (cf. Traykovski et al., 2000). The high frequency acoustic pulse cannot penetrate consolidating muddy bed or sandy bed below fluid mud. Based on this, the thickness of the layer was estimated to approximate between 10 and 15 cm (Figure 9e-g, shaded area). The high value of the SSC disappeared during the storm peak and the peak of the SSC shifted upward, suggesting the layer was vertically diffused due to high vertical velocity. During the waning phase of the storm, the fluid mud layer was re-established. However, the layer is highly unstable according to the SSC profile during the deployment (not in the figure). It is implied that this unstable

layer, which can easily be re-suspended, could not be settled long enough to become consolidated, because of the frequency of the winter storms. Allison et al. (2000) demonstrated that sediment cores taken on the southwest of Ship Shoal had no modern sedimentary records, implying these muddy sediments on the sandy relict sediments cannot become consolidated hard mud bed without leaving any modern sediment records as reported by the recent core surveys at the southwest of the shoal.

There are few data available to prove this hypothesis. It is still unclear as to the fate of this transient mud sediment covering the sandy bottom. Surveys will be undertaken to examine bottom sediments using box cores and underwater cameras to better understand mud distribution during various seasons. Also, the ongoing modeling studies of waves, currents and cohesive sediment transport should provide a more comprehensive analysis of the shoal and its unique processes, which are important to the physical and biological processes of the study area, and for future sand dredging activities.

## **CONCLUSIONS**

To examine wave-climate and wave-bottom interactions over a heterogeneous shoal, field surveys were conducted and the following results were obtained. (I) Bottom sediments on the shoal vary substantially depending on fluvial sediment inputs and winter storms; (II) The north-south trend in current direction during storms suggests a dominant sediment transport direction during the deployment; (III) Rates of transport for cohesive sediment is much higher than those numerically derived non-cohesive sediment transport during the deployment; (IV) A fluid mud layer is formed during fair weather on the shoal and is re-suspended, mixed and re-deposited during storms, implying fine-grained sediment are ephemeral resulting in no sedimentary records on the deposited sediments previously reported by Allison et al. (2000).

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